Enhancing Digital Connectivity at LNBTI: A Proposal for Campus LAN Modernization

A Project Proposal By:

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Introduction

The Lanka Nippon BizTech Institute (LNBTI) Main Building currently lacks a structured and efficient Local Area Network (LAN), which hinders smooth digital communication, resource sharing, and access to essential academic and administrative systems. In today's technology-driven academic environment, a strong and reliable LAN is critical for supporting teaching, learning, research, and institutional operations. This proposal aims to design and implement a robust network infrastructure tailored to the needs of LNBTI, ensuring seamless connectivity, improved performance, and scalability for future growth. The project involves conducting background research, identifying appropriate equipment, estimating costs, and designing an optimal network layout for the entire building.

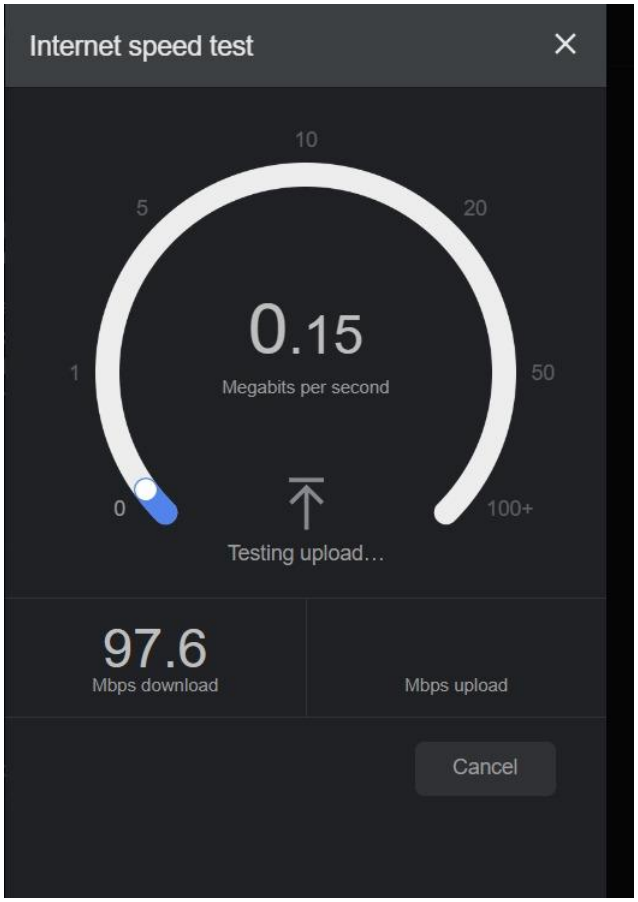
Objectives of the Proposal

1. To set up a reliable and fast Local Area Network (LAN) for the LNBTI Main Building to support daily academic and administrative activities.
2. To connect all computers, printers, and other devices so that staff and students can easily share files and access online resources.
3. To choose the right network equipment such as switches, routers, access points, and cables to ensure full coverage and smooth performance.
4. To provide a clear cost estimate that includes all equipment, installation and student and staff training sessions.
5. To create a proper IP address plan using subnetting to manage network connections efficiently and securely.
6. To make sure the new network is safe, easy to manage, and ready for future expansion as the institute grows.
7. To improve communication, learning, and overall productivity by upgrading the IT infrastructure of the building.

Current Network Situation at LNBTI

Internet Connection:

- Provided by SLT, with a 2 TB/month data cap.
- Current bandwidth is ~97 Mbps download / 0.05 Mbps upload, based on speed test. – Below is the speed test run at morning (8.30AM) when the data traffic is usually minimum.



- Upload bandwidth is severely limited, affecting file submissions and cloud backups.

Firewall and Security:

- Existing firewall infrastructure is outdated.
- No application-aware traffic control or bandwidth shaping.
- No effective control over student vs. staff usage beyond simple VLAN separation.

Network Segmentation:

- Network is split into two broad segments: Academic Staff and Students.
- No detailed traffic prioritization (e.g., exams, board meetings).
- Exam network gets full priority, which causes others to lose connectivity during exam time.

Access Point Infrastructure:

- Existing Wi-Fi APs are older models (likely Wi-Fi 4 or Wi-Fi 5).
- Each AP supports only 10 to 25 users stably before severe performance drop.
- Many floors do not have enough APs to support actual user density (especially in Basement, 2nd, and 3rd floors).

Network Stability & Coverage Issues:

- Many users report inability to upload files or maintain stable Zoom calls.
- Coverage is incomplete in lecture rooms and self-study areas.

User Limitations:

- Students have a 500 MB daily quota, after which speed is throttled.
- Legitimate updates (OS, software) often consume the quota unknowingly.
- No per-user bandwidth shaping or quota transparency.

Wired Infrastructure:

- Most areas rely on Wi-Fi, even for exam rooms and labs.
- Computer Lab PCs are wirelessly connected, leading to instability during online exams.

Server Room:

- Located on the 2nd floor.
- Current setup lacks redundancy, dedicated VLANs, or proper uptime monitoring.

Scalability:

- The current setup does not scale well with growing student numbers or device usage.
- Internet data cap is exceeded frequently, leading to service degradation.

Proposed Network Solution Overview

To address the growing demands and connectivity issues at LNBTI, we propose a scalable and future-proof LAN infrastructure upgrade. This design ensures reliable access, higher speeds, better control, and secure segmentation for staff, students, and critical systems.

1. Internet Bandwidth Upgrade

- Upgrade: From the current 2 TB/month capped broadband connection to a dedicated 2.5 TB/month leased line from SLT.
- Speed: Target 1 Gbps symmetrical (upload/download).
- Justification:
    - Supports high-bandwidth usage such as:
        - Online examinations
        - Video conferencing (Zoom, Google Meet)
        - Cloud-based platforms (e.g., Moodle, Contentful, GitHub)
        - Regular web access, streaming, and software updates
    - Ensures stable connectivity for over 500 concurrent users across all five floors.
    - Reduces latency and jitter, improving VoIP and real-time class streaming quality.
    - Guarantees uptime and SLAs better suited for an educational institute.

2. Core Network Upgrade

- Core Device: A central Layer 3 switch (Cisco Catalyst 3560 or equivalent).
- Purpose: Enables inter-VLAN routing, network segmentation, and central management.
- Justification:
    - Facilitates VLAN-based segregation of traffic (e.g., Staff, Student, Administration, Servers).
    - Handles routing between subnets locally, improving internal communication speed.

3. Floor-Level Switching

- Access Layer Devices: Each floor is equipped with Layer 2 PoE switches (Cisco 2960 series or equivalent).
- Justification:
    - Powers access points and VoIP devices without the need for separate adapters.
    - Handles all endpoint connections on each floor, providing up to 48 ports.
    - Connects to the core switch via trunk links for VLAN propagation.

4. Wi-Fi 6 Access Point Deployment

- AP Model: Aruba Instant On AP22 (Wi-Fi 6)
- Placement: Strategically deployed across all floors and key rooms.
- Justification:
  - Replaces legacy Wi-Fi 4 APs which limit to ~50 users/AP and suffer under load.
  - Wi-Fi 6 supports more simultaneous users, better performance in congested environments, and higher speeds.
  - Enables future-ready wireless infrastructure with OFDMA, MU-MIMO, and low latency.

5. Structured Cabling

- Type: Cat6A structured cabling (~5,500 meters total).
- Justification:
  - Supports 10 Gbps data rates.
  - Improved shielding reduces interference, ensuring consistent performance for APs, PCs, and printers.

6. Server Room Equipment

- Router + Firewall: 1 Gbps router with NGFW (FortiGate/Sophos) and 3-year UTM license.
- Rack: 42U server rack with cable management and PDUs.

- Power Backup: 5 kVA online UPS to ensure uninterrupted server operation.

7. Security and Segmentation

- VLAN Assignment: Devices grouped by role (e.g., Staff VLAN, Student VLAN, Server VLAN).
- Benefits:
    - Enhanced security and access control.
    - Improved bandwidth allocation per department.
    - Easier troubleshooting and monitoring.

8. Scalability and Monitoring

- Designed to scale up for future users, devices, and services.
- Includes a network monitoring system (with SNMP and RADIUS) to:
    - Track AP usage
    - Monitor switch performance
    - Enforce authentication

9. Training and Maintenance

- Staff and student ambassadors to receive network operation training.
- Design includes documentation, labeling, and support contracts for all critical components.

Question 01 - Equipment List

<table>
<tr><td rowspan="9">LNBTI Main Campus
Router x 1
L3 Core Switch x 1</td><td rowspan="5">Basement</td><td rowspan="5">Layer 2 PoE switch x 1
Multi-function printer x 1</td><td>Lecture Room 05</td><td colspan="2">Wi-Fi 6 Access Point x 1
PC x 1</td></tr>
<tr><td>Lecture Room 06</td><td>PC x 1</td><td rowspan="3">Wi-Fi 6 Access Point x 1</td></tr>
<tr><td>Lecture Room 07</td><td>PC x 1</td></tr>
<tr><td>Counselling Room</td><td>PC x 1</td></tr>
<tr><td></td></tr>
<tr><td rowspan="4"></td><td rowspan="4">Ground Floor</td><td rowspan="4">Layer 2 PoE switch x 1
Multi-Function printer x 3</td><td>Reception</td><td colspan="2">Wi-Fi 4 Access Point x 1
PC x 1
QR code scanner</td></tr>
<tr><td>Registrar Room</td><td>PC x 2</td><td rowspan="6">Wi-Fi 6 Access Point x 2</td></tr>
<tr><td>Admission Room</td><td>PC x 5</td></tr>
<tr><td>Administration Room</td><td>PC x 2</td></tr>
<tr><td></td></tr>
<tr><td></td><td></td><td></td><td>Examination Division</td><td>PC x 1</td></tr>
<tr><td></td><td></td><td></td><td>Staff Room</td><td>PC x 30</td></tr>
<tr><td></td><td></td><td></td><td>Student Affairs</td><td>PC x 3</td></tr>
</table>

	First Floor	Layer 2 PoE switch x 1	Cafeteria	Wi-Fi 6 Access Point x 2	
	Second Floor	Layer 2 PoE switch x 1 Multi-function printer x 1	Self-Study Room	Wi-Fi 6 Access Point x 1	
			Lecture Room 02	PC x 1	Wi-Fi 6 Access Point x 2
			Lecture Room 04		
			Server Room	Router x 1 Layer 3 switch x 1	
			Board Room	Wi-Fi 6 Access Point x 1 PC x 1	
	Third Floor	Layer 2 PoE switch x 1	Computer Lab	PC x 40 Wi-Fi 6 Access Point x 2	
			Library	PC x 1	Wi-Fi 6 Access Point x 1
			Center for QA	PC x 1 Multi-function printer x 1	
			Research Lab	PC x 5 Wi-Fi 6 Access Point x 1	

Question 02 - Cost Estimation

Item	Qty.	Unit Price (Rs.)	Total Price (Rs.)
Wi-Fi 6 Access Point	13	185,000.00	2,405,000.00
Wi-Fi 4 Access Point	01	55,000.00	55,000.00
Layer 2 PoE Switch	05	250,000.00	1,250,000.00
Layer 3 Switch (Core)	01	600,000.00	600,000.00
Router (Edge Firewall)	01	700,000.00	700,000.00
Multi-Function Printer	06	150,000.00	900,000.00
QR Code Scanner	01	20,000.00	20,000.00
Structured Cabling & Installation		800,000.00	800,000.00
Training Sessions		100,000.00	100,000.00
Professional Services (Config & Testing)		300,000.00	300,000.00
Contingency (approx. 5%)		250,000.00	250,000.00
Total Estimated Rs.7,380,000.00			

Justification for The Above Choices

1. Router (Edge Firewall – 1 Unit)

- Only one enterprise-grade router is proposed because it will act as the central gateway between the Internet and the internal network. It will:
- Perform NAT (Network Address Translation) for all internal IPs,
- Enforce security policies and access control between VLANs,
- Handle traffic shaping and Quality of Service (QoS) to prioritize exams, video conferencing, and academic traffic,
- Support VPN access and future remote management needs,
- Ensure policy-based routing, if the Internet is later split into student/staff lines.
- One router is sufficient for a medium-sized campus if it is properly sized (e.g., MikroTik CCR2004 or FortiGate 60F), supporting 1 Gbps throughput or more.

2. Layer 3 Switch (Core – 1 Unit)

- The Layer 3 switch will serve as the central core switch located in the server room. It is essential because it:
- Handles inter-VLAN routing between student, staff, exam, and server networks,
- Supports high-speed 1 Gbps or 10 Gbps uplinks to floor switches,

- Offloads internal routing from the router, reducing latency,
- Allows creation of custom subnet policies for traffic isolation and management.
- This setup allows scalable segmentation of network traffic by department, function (exam/staff/IoT), or access level.

3. Layer 2 PoE Switches (5 Units, One per Floor)

- Each floor is equipped with a dedicated Layer 2 switch to:
- Provide wired network access to PCs, printers, and access points,
- Supply PoE (Power over Ethernet) to Wi-Fi 6 APs and multifunction printers, reducing the need for extra power adapters,
- Connect to the core switch via uplink ports for seamless integration.
- This ensures simple, manageable network expansion per floor while maintaining centralized control.

4. Wi-Fi 6 Access Points (13 Units)

- The Wi-Fi 6 APs (e.g., Aruba Instant On AP22) have been selected to address:
- Overloaded and outdated Wi-Fi 4 APs, which currently cannot handle modern traffic loads,
- The need for dual-band (2.4 GHz + 5 GHz) performance for high-density user areas,
- Reliable wireless connectivity for students, staff, and guest devices across lecture halls, labs, boardrooms, and open spaces,

- Support for modern protocols (e.g., MU-MIMO, OFDMA, WPA3) and better handover between rooms.
- The total of 13 APs was calculated based on adjusted room layout and usage density, with APs strategically placed to cover grouped rooms where appropriate.

5. Multi-Function Printers (6 Units)

- Rather than separate printers and scanners, the proposal includes multi-function printers (MFPs) that offer:
- Printing, scanning, and copying in one compact unit,
- Network connectivity to allow access from any authorized PC on the same VLAN,
- Cost-efficiency and ease of maintenance compared to multiple separate devices,
- Placement in key functional zones (basement, ground floor, 2nd floor, QA office, etc.) to minimize congestion and travel distance for users.
- This reduces device clutter and supports digital document handling across campus.

6. QR Code Scanner (1 Unit)

- A single QR scanner is proposed for the reception or entry checkpoint to support:
- Fast student ID verification, attendance tracking, or access control integration,

- Digital transformation of manual entry logs (optional integration with future systems).

7. Structured Cabling & Installation

- Reliable wired connectivity is essential for:
- Stable exam room PCs and lab environments,
- Uplink connections between switches and the core,
- Reducing wireless congestion by offloading fixed-position devices (e.g., PCs and printers).
- Cat6 cabling with structured layouts, trunking, and patch panels is included to ensure high throughput and long-term durability.

8. Training Sessions

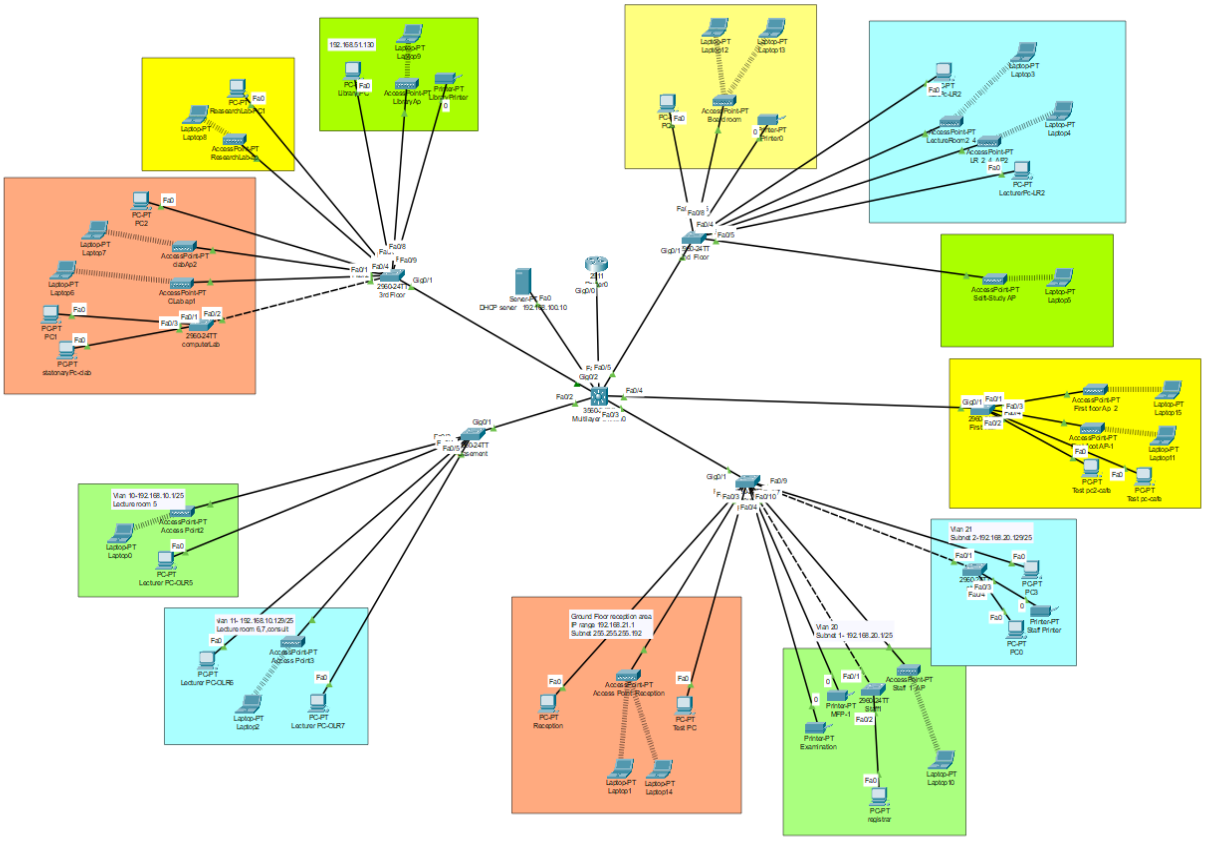
- To ensure smooth adoption and maintenance:
- IT staff will be trained on managing VLANs, APs, and monitoring tools,
- Academic staff and student ambassadors will be trained on how to use the new systems and report network issues.

9. Professional Services & Contingency

- Specialist services for:
- Design, configuration, testing, and documentation of the network,

10.Contingency budget covers unexpected on-site adjustments, extra cables, wall brackets, or licensing.

Question 03 – Network Design



VLAN	Location / Purpose	Network/Subnet	Subnet Mask	Gateway IP	Start IP (DHCP)	Max Hosts
VLAN 10	Lecture Room 5	192.168.10.0/25	255.255.255.128	192.168.10.1	192.168.10.2	126
VLAN 11	Lecture Rooms 6, 7 and Consult Room	192.168.10.128/25	255.255.255.128	192.168.10.129	192.168.10.130	126
VLAN 20	Registrar, Admin, Admission, Examinatio n (Staff 1)	192.168.20.0/25	255.255.255.128	192.168.20.1	192.168.20.15	126
VLAN 21	Staff Room & Student Affairs (Staff 2)	192.168.20.128/25	255.255.255.128	192.168.20.129	192.168.20.141	126
VLAN 22	Ground Floor Reception Area	192.168.21.0/25	255.255.255.128	192.168.21.1	192.168.21.2	126
VLAN 30	1st Floor Cafeteria	192.168.30.0/25	255.255.255.128	192.168.30.1	192.168.30.2	100
VLAN 40	Lecture Rooms 2 and 4 (2nd Floor)	192.168.40.0/25	255.255.255.128	192.168.40.1	192.168.40.2	126
VLAN 41	Self-Study Area (2nd Floor)	192.168.41.0/25	255.255.255.128	192.168.41.1	192.168.41.2	126
VLAN 42	Board Room (2nd Floor)	192.168.42.0/26	255.255.255.192	192.168.42.1	192.168.42.2	61
VLAN 50	Computer Lab (3rd Floor)	192.168.50.0/24	255.255.255.0	192.168.50.1	192.168.50.51	254
VLAN 51	Research Lab (3rd Floor)	192.168.51.0/25	255.255.255.128	192.168.51.1	192.168.51.2	126
VLAN 52	Library + QA (3rd Floor)	192.168.51.128/25	255.255.255.128	192.168.51.129	192.168.51.140	126

Conclusion

The proposed network upgrade at LNBTI addresses the pressing issues of limited bandwidth, unreliable wireless coverage, and lack of scalability that currently hinder academic and administrative operations. By implementing a modern LAN infrastructure—including a 1 Gbps symmetrical leased line, enterprise-grade Layer 3 core switching, floor-level PoE switches, and Wi-Fi 6 access points—the institution will experience a significant improvement in network speed, reliability, and user capacity.

Structured cabling, VLAN-based segmentation, and network monitoring tools will ensure high performance, security, and maintainability. The investment of Rs. 7.38 million is strategically allocated, with component selection and cost estimation based on reliable market sources. This upgrade empowers LNBTI to support digital learning platforms, video conferencing, online assessments, and administrative workflows without disruption, while being well-prepared for future growth.

References

1. **Speed Test Data** – Real-time bandwidth evaluation using Speedtest by Ookla
2. **Network Equipment Pricing** – Component cost estimates referenced from Richcom.lk